

Simplex II: Cycling and Anticycling

How can simplex keep changing bases without moving—and how can we prevent a cycle?

3.4 · Cycling and anticycling

3D · Comparing anticycling rules

Finite termination and computational effort

Appendix · Complete 3D calculations, practice, and proof

Lecture 6 · Bertsimas–Tsitsiklis §3.4

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Can Simplex Cycle? Example 3.6

Simplex I showed a zero-length pivot: the basis changed while the point stayed fixed. Can such pivots return to the starting basis?

$$\begin{aligned} \min \quad & f(x) = -\frac{3}{4}x_1 + 20x_2 - \frac{7}{2}x_3 + 6x_4 - 3x_7 \\ \text{s.t.} \quad & \frac{1}{4}x_1 - 8x_2 - x_3 + 9x_4 + x_5 = 0, \\ & \frac{1}{2}x_1 - 12x_2 - \frac{1}{2}x_3 + 3x_4 + x_6 = 0, \\ & x_3 + x_7 = 1, \quad x_1, \dots, x_7 \geq 0. \end{aligned}$$

Start with $\mathcal{B} = (x_5, x_6, x_7)$: $x = (0, 0, 0, 0, 0, 0, 1)$, $f_B = -3$.

First write a dictionary, just as in Simplex I.

Example 3.6: Start with a Dictionary

Basic variables: (x_5, x_6, x_7) . Nonbasic variables: (x_1, x_2, x_3, x_4) .

$$\begin{aligned}x_5 &= -\frac{1}{4}x_1 + 8x_2 + x_3 - 9x_4 \\x_6 &= -\frac{1}{2}x_1 + 12x_2 + \frac{1}{2}x_3 - 3x_4 \\x_7 &= 1 - x_3\end{aligned}$$

Substitute $x_7 = 1 - x_3$ into the minimization objective:

$$\begin{aligned}f &= -\frac{3}{4}x_1 + 20x_2 - \frac{7}{2}x_3 + 6x_4 - 3(1 - x_3) \\&= -3 - \frac{3}{4}x_1 + 20x_2 - \frac{1}{2}x_3 + 6x_4.\end{aligned}$$

Set the nonbasic variables to zero: $(x_5, x_6, x_7) = (0, 0, 1)$ and $f = -3$.

The objective coefficients are the reduced costs for this basis.

From Dictionary Equations to a Tableau

A **tableau** records each equation’s right-hand side and left-hand coefficients.

Constraint rows: move the nonbasic terms to the left. For example,

$$\frac{1}{4}x_1 - 8x_2 - x_3 + 9x_4 + x_5 = 0.$$

Objective row R_0 : move f left and the constant right:

$$-f - \frac{3}{4}x_1 + 20x_2 - \frac{1}{2}x_3 + 6x_4 = 3.$$

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$-\frac{3}{4}$	20	$-\frac{1}{2}$	6	0	0	0
x_5	0	$\frac{1}{4}$	−8	−1	9	1	0	0
x_6	0	$\frac{1}{2}$	−12	$-\frac{1}{2}$	3	0	1	0
x_7	1	0	0	1	0	0	0	1

R_0 omits $-f$. Column 0 is $-f_B = 3$; the other entries are unchanged reduced costs.

Example 3.6: Pivot 1

Most-negative reduced cost enters; smallest basic-variable index breaks a minimum-ratio tie.

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$-\frac{3}{4}$	20	$-\frac{1}{2}$	6	0	0	0
x_5	0	$\frac{1}{4}$	-8	-1	9	1	0	0
x_6	0	$\frac{1}{2}$	-12	$-\frac{1}{2}$	3	0	1	0
x_7	1	0	0	1	0	0	0	1

$$\bar{c}_1 = -\frac{3}{4} < -\frac{1}{2} = \bar{c}_3, \quad \theta^* = \min \left\{ \frac{0}{1/4}, \frac{0}{1/2} \right\} = 0.$$

x_5 and x_6 tie at zero; the smaller index makes x_5 leave.

Example 3.6: Pivot 2

After pivot 1: x_1 entered, x_5 left. Same point, same $f = -3$. Which pivot comes next?

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	0	-4	$-\frac{7}{2}$	33	3	0	0
x_1	0	1	-32	-4	36	4	0	0
x_6	0	0	4	$\frac{3}{2}$	-15	-2	1	0
x_7	1	0	0	1	0	0	0	1

$$\bar{c}_2 = -4 < -\frac{7}{2} = \bar{c}_3, \quad \theta^* = \frac{0}{4} = 0.$$

Only x_6 has a positive entry in column x_2 ; it leaves at step zero.

Example 3.6: Pivot 3

After pivot 2: x_2 entered, x_6 left. Same point, same $f = -3$. Which pivot comes next?

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	0	0	-2	18	1	1	0
x_1	0	1	0	8	-84	-12	8	0
x_2	0	0	1	$\frac{3}{8}$	$-\frac{15}{4}$	$-\frac{1}{2}$	$\frac{1}{4}$	0
x_7	1	0	0	1	0	0	0	1

$$\bar{c}_3 = -2, \quad \theta^* = \min \left\{ \frac{0}{8}, \frac{0}{3/8}, \frac{1}{1} \right\} = 0.$$

x_1 and x_2 tie at zero; the smaller index makes x_1 leave.

Example 3.6: Pivot 4

After pivot 3: x_3 entered, x_1 left. Same point, same $f = -3$. Which pivot comes next?

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$\frac{1}{4}$	0	0	-3	-2	3	0
x_3	0	$\frac{1}{8}$	0	1	$-\frac{21}{2}$	$-\frac{3}{2}$	1	0
x_2	0	$-\frac{3}{64}$	1	0	$\frac{3}{16}$	$\frac{1}{16}$	$-\frac{1}{8}$	0
x_7	1	$-\frac{1}{8}$	0	0	$\frac{21}{2}$	$\frac{3}{2}$	-1	1

$$\bar{c}_4 = -3 < -2 = \bar{c}_5, \quad \theta^* = \min \left\{ \frac{0}{3/16}, \frac{1}{21/2} \right\} = 0.$$

The x_2 row limits the step at zero; the x_7 row would allow $2/21$.

Example 3.6: Pivot 5

After pivot 4: x_4 entered, x_2 left. Same point, same $f = -3$. Which pivot comes next?

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$-\frac{1}{2}$	16	0	0	-1	1	0
x_3	0	$-\frac{5}{2}$	56	1	0	2	-6	0
x_4	0	$-\frac{1}{4}$	$\frac{16}{3}$	0	1	$\frac{1}{3}$	$-\frac{2}{3}$	0
x_7	1	$\frac{5}{2}$	-56	0	0	-2	6	1

$$\bar{c}_5 = -1 < -\frac{1}{2} = \bar{c}_1, \quad \theta^* = \min \left\{ \frac{0}{2}, \frac{0}{1/3} \right\} = 0.$$

x_3 and x_4 tie at zero; the smaller index makes x_3 leave.

Example 3.6: Pivot 6

After pivot 5: x_5 entered, x_3 left. Same point, same $f = -3$. Which pivot comes next?

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$-\frac{7}{4}$	44	$\frac{1}{2}$	0	0	-2	0
x_5	0	$-\frac{5}{4}$	28	$\frac{1}{2}$	0	1	-3	0
x_4	0	$\frac{1}{6}$	-4	$-\frac{1}{6}$	1	0	$\frac{1}{3}$	0
x_7	1	0	0	1	0	0	0	1

$$\bar{c}_6 = -2 < -\frac{7}{4} = \bar{c}_1, \quad \theta^* = \frac{0}{1/3} = 0.$$

Only x_4 has a positive entry in column x_6 ; it leaves at step zero.

Example 3.6: After Pivot 6

x_6 entered and x_4 left with $\theta^* = 0$. Compare the result with the initial tableau.

After pivot 6 · the initial tableau returns

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$-\frac{3}{4}$	20	$-\frac{1}{2}$	6	0	0	0
x_5	0	$\frac{1}{4}$	-8	-1	9	1	0	0
x_6	0	$\frac{1}{2}$	-12	$-\frac{1}{2}$	3	0	1	0
x_7	1	0	0	1	0	0	0	1

Initial tableau restored. Basis (x_5, x_6, x_7) , objective $f = -3$, point $x = (0, 0, 0, 0, 0, 0, 1)$.

The same entering and leaving rules now select the same first pivot. The sequence repeats indefinitely:
simplex cycles.

Six Pivots Return to the Same Basis

$$\begin{aligned}(x_5, x_6, x_7) &\rightarrow (x_1, x_6, x_7) \rightarrow (x_1, x_2, x_7) \\ &\rightarrow (x_3, x_2, x_7) \rightarrow (x_3, x_4, x_7) \\ &\rightarrow (x_5, x_4, x_7) \rightarrow (x_5, x_6, x_7).\end{aligned}$$

Every step has $\theta^* = 0$. Every tableau represents the same point and objective $f = -3$.

The sixth tableau equals the first. The same rules now make the same choices again: the algorithm cycles indefinitely.

A finite number of bases is not enough: we must also prevent repeated bases.

What an Anticycling Rule Must Achieve

Degeneracy may prevent strict objective improvement. An anticycling rule supplies another reason that a basis cannot recur.

We will study lexicographic pivoting and Bland's rule. Both allow degenerate pivots and guarantee finite termination from a feasible basis.

A consequence

For a feasible standard-form LP with finite optimal value, an optimal basis exists:

$$B^{-1}b \geq 0, \quad \bar{c}^\top = c^\top - c_{\mathcal{B}}^\top B^{-1}A \geq 0.$$

We next learn the rules; the following section supplies an initial feasible basis.

Lexicographic Order: Use the First Difference

Definition 3.5

For distinct vectors r, s , compare their entries from left to right. Write $r >_L s$ if the first unequal entry is larger in r ; write $r <_L s$ if it is smaller.

$$(0, 2, \boxed{3}, 0) >_L (0, 2, \boxed{1}, 4).$$

The first two entries tie; the third entry decides. The final entry does not matter.

$$(\boxed{0}, 4, 5, 0) <_L (\boxed{1}, 2, 1, 2).$$

Here the first entry decides, even though later entries go the other way.

The Lexicographic Pivoting Rule

1 • Choose an improving column

Choose any nonbasic x_j with $\bar{c}_j < 0$. Its entries are $t_{ij} = (B^{-1}A_j)_i$.

2 • Compare normalized eligible rows

Let R_i be the full tableau row: RHS, then coefficients in a fixed column order. For each $t_{ij} > 0$, divide R_i by t_{ij} and choose the lexicographically smallest result:

$$\frac{R_\ell}{t_{\ell j}} <_L \frac{R_i}{t_{ij}} \quad (i \neq \ell, t_{ij} > 0).$$

Then $x_{B(\ell)}$ leaves. Column 0 first minimizes the ordinary ratio; later columns resolve a tie. If no row is eligible, the improving direction is unbounded.

Example 3.6 Again: Keep the Usual Column Order

Restart the same **minimization** problem at $\mathcal{B} = (x_5, x_6, x_7)$. Keep the most-negative reduced-cost entering rule.

$$\begin{aligned}x_5 &= -\frac{1}{4}x_1 + 8x_2 + x_3 - 9x_4 \\x_6 &= -\frac{1}{2}x_1 + 12x_2 + \frac{1}{2}x_3 - 3x_4 \\x_7 &= 1 - x_3 \\f &= -3 - \frac{3}{4}x_1 + 20x_2 - \frac{1}{2}x_3 + 6x_4\end{aligned}$$

Compare tableau rows from left to right, always in this order:

$$\text{RHS} \mid x_1, x_2, x_3, x_4, x_5, x_6, x_7.$$

Lex Pivot 1: The First Tie Still Chooses x_5

x_1 enters: $-\frac{3}{4} < -\frac{1}{2}$. Rows x_5, x_6 tie at ratio zero; the x_7 row is ineligible.

R_{x_i} is the row for basic x_i ; divide by its x_1 -column entry.

row	RHS	x_1	x_2	x_3	x_4	x_5	x_6	x_7
$R_{x_5}/(\frac{1}{4})$	0	1	-32	-4	36	4	0	0
$R_{x_6}/(\frac{1}{2})$	0	1	-24	-1	6	0	2	0

The RHS and x_1 entries tie. The first difference is in column x_2 :

$$-32 < -24 \implies \frac{R_{x_5}}{1/4} <_L \frac{R_{x_6}}{1/2}.$$

x_5 **leaves** at $\theta = 0$. This first pivot agrees with the earlier cycling run.

After Pivot 1: Choose the Second Pivot

Solve the old x_5 equation for x_1 and substitute. The basis becomes (x_1, x_6, x_7) :

$$x_1 = 32x_2 + 4x_3 - 36x_4 - 4x_5$$

$$x_6 = -4x_2 - \frac{3}{2}x_3 + 15x_4 + 2x_5$$

$$x_7 = 1 - x_3$$

$$f = -3 - 4x_2 - \frac{7}{2}x_3 + 33x_4 + 3x_5$$

x_2 **enters:** $-4 < -\frac{7}{2}$. With the other nonbasic variables zero, $x_6 = -4x_2 \geq 0$ forces $x_2 = 0$.

Pivot 2: x_6 **leaves,** $\theta = 0$. No ratio tie occurs. The first two pivots still match the cycling run.

After Pivot 2: A New Tie to Resolve

Solve the old x_6 equation for x_2 , then substitute. Now $\mathcal{B} = (x_1, x_2, x_7)$:

$$x_1 = -8x_3 + 84x_4 + 12x_5 - 8x_6$$

$$x_2 = -\frac{3}{8}x_3 + \frac{15}{4}x_4 + \frac{1}{2}x_5 - \frac{1}{4}x_6$$

$$x_7 = 1 - x_3$$

$$f = -3 - 2x_3 + 18x_4 + x_5 + x_6$$

x_3 **enters:** $\bar{c}_3 = -2$ is the only negative reduced cost.

$$\theta^* = \min \left\{ \frac{0}{8}, \frac{0}{3/8}, \frac{1}{1} \right\} = 0.$$

x_1 and x_2 tie at zero. Which complete normalized row is smaller?

Lex Pivot 3: This Choice Breaks the Cycle

Compare the eligible rows in the same order: RHS, then $x_1, \dots, x_7$.

R_{x_i} is the row for basic x_i ; divide by its x_3 -column entry.

row	RHS	x_1	x_2	x_3	x_4	x_5	x_6	x_7
$R_{x_1}/(8)$	0	$\frac{1}{8}$	0	1	$-\frac{21}{2}$	$-\frac{3}{2}$	1	0
$R_{x_2}/(\frac{3}{8})$	0	0	$\frac{8}{3}$	1	-10	$-\frac{4}{3}$	$\frac{2}{3}$	0
$R_{x_7}/(1)$	1	0	0	1	0	0	0	1

The x_7 row has ratio 1. Among the zero-ratio rows, column x_1 decides:

$$0 < \frac{1}{8} \implies \frac{R_{x_2}}{3/8} <_L \frac{R_{x_1}}{8}.$$

x_2 **leaves at** $\theta = 0$. The cycling run instead made x_1 leave. We now reach a different basis.

After Pivot 3: The Fourth Pivot Can Move

Pivot on the old x_2 row to make x_3 basic. The new dictionary is

$$\begin{aligned}x_1 &= \frac{64}{3}x_2 + 4x_4 + \frac{4}{3}x_5 - \frac{8}{3}x_6 \\x_3 &= -\frac{8}{3}x_2 + 10x_4 + \frac{4}{3}x_5 - \frac{2}{3}x_6 \\x_7 &= 1 + \frac{8}{3}x_2 - 10x_4 - \frac{4}{3}x_5 + \frac{2}{3}x_6 \\f &= -3 + \frac{16}{3}x_2 - 2x_4 - \frac{5}{3}x_5 + \frac{7}{3}x_6\end{aligned}$$

x_4 **enters:** $-2 < -\frac{5}{3}$. Set $x_4 = \theta$, with $x_2 = x_5 = x_6 = 0$:

$$x_1 = 4\theta, \quad x_3 = 10\theta, \quad x_7 = 1 - 10\theta.$$

Pivot 4: x_7 **leaves** at $\theta = \frac{1}{10}$. Only x_7 decreases; the objective drops to $-\frac{16}{5}$.

After Pivot 4: One More Improving Pivot

Solve the old x_7 equation for x_4 and substitute. Now $\mathcal{B} = (x_1, x_3, x_4)$:

$$x_1 = \frac{2}{5} + \frac{112}{5}x_2 + \frac{4}{5}x_5 - \frac{12}{5}x_6 - \frac{2}{5}x_7$$

$$x_3 = 1 - x_7$$

$$x_4 = \frac{1}{10} + \frac{4}{15}x_2 - \frac{2}{15}x_5 + \frac{1}{15}x_6 - \frac{1}{10}x_7$$

$$f = -\frac{16}{5} + \frac{24}{5}x_2 - \frac{7}{5}x_5 + \frac{11}{5}x_6 + \frac{1}{5}x_7$$

x_5 **enters:** $-\frac{7}{5}$ is the only negative reduced cost. Only x_4 decreases:

$$x_4 = \frac{1}{10} - \frac{2}{15}\theta \geq 0 \quad \Rightarrow \quad \theta \leq \frac{3}{4}.$$

Pivot 5: x_4 **leaves at** $\theta = \frac{3}{4}$. Solve for x_5 and substitute once more.

Example 3.6: Optimal After Five Lex Pivots

The final basis is (x_1, x_3, x_5) ; the nonbasic variables are (x_2, x_4, x_6, x_7) .

$$x_1 = 1 + 24x_2 - 6x_4 - 2x_6 - x_7$$

$$x_3 = 1 - x_7$$

$$x_5 = \frac{3}{4} + 2x_2 - \frac{15}{2}x_4 + \frac{1}{2}x_6 - \frac{3}{4}x_7$$

$$f = -\frac{17}{4} + 2x_2 + \frac{21}{2}x_4 + \frac{3}{2}x_6 + \frac{5}{4}x_7$$

All nonbasic reduced costs are positive. Set the nonbasic variables to zero:

$$x^* = (1, 0, 1, 0, \frac{3}{4}, 0, 0), \quad f^* = -\frac{17}{4}.$$

Three zero-length pivots, then two positive steps. The changed leaving choice at pivot 3 prevents the earlier six-pivot cycle.

Why the Lexicographic Leaving Choice Is Unique

Assume two different eligible rows tie in every entry after normalization.

$$\frac{R_i}{t_{ij}} = \frac{R_k}{t_{kj}} \implies R_i = \frac{t_{ij}}{t_{kj}} R_k.$$

The complete rows would be proportional.

Deleting column 0 leaves proportional rows of $T = B^{-1}A$, so T would not have independent rows.

$$\text{rank}(B^{-1}A) = \text{rank}(A) = m.$$

This contradicts the full-row-rank assumption. Distinct eligible rows therefore cannot tie lexicographically.

What Guarantees Termination More Generally?

Example 3.6 reached an optimal dictionary after five pivots. We verified this run directly.

Computing the pivots

Normalize eligible rows and compare them lexicographically. No extra positivity test is used to choose the leaving row.

Guaranteeing termination

Theorem 3.4 uses a property called **lex positivity**. Our example's initial tableau already has it.

We now define this property and explain its role in the no-cycling guarantee.

Lex Positivity: A Condition for the Termination Theorem

A vector is **lexicographically positive** if its first nonzero entry is positive: $r >_L 0$.

Example 3.6 already starts this way. In the order RHS, $x_1, \dots, x_7$, its constraint rows begin

$$R_{x_5} = (0, \boxed{\frac{1}{4}}, -8, \dots),$$

$$R_{x_6} = (0, \boxed{\frac{1}{2}}, -12, \dots),$$

$$R_{x_7} = (\boxed{1}, 0, 0, \dots).$$

All three rows are lex positive. Later entries may be negative; only the first nonzero entry decides. The zero vector is not lex positive.

Theorem 3.4: Finite Termination of Lexicographic Simplex

Consider $\min\{c^\top x : Ax = b, x \geq 0\}$, with $\text{rank}(A) = m$.

Start from a feasible basis whose constraint rows are lexicographically positive. Keep the column order fixed and use the lexicographic pivoting rule.

Then all of the following hold:

- (a) Every constraint row remains lexicographically positive at every iteration.
- (b) The complete objective row $R_0 = [-f_B \mid \bar{c}^\top]$ strictly increases lexicographically at every pivot.
- (c) The algorithm terminates after finitely many pivots, producing an optimal basis or certifying that the objective is unbounded below.

Intuition: Before Pivot 3

Example 3.6 after two pivots: $\mathcal{B} = (x_1, x_2, x_7)$.

$$\begin{aligned} x_1 &= -8x_3 + 84x_4 + 12x_5 - 8x_6 \\ x_2 &= -\frac{3}{8}x_3 + \frac{15}{4}x_4 + \frac{1}{2}x_5 - \frac{1}{4}x_6 \\ x_7 &= 1 - x_3 \\ f &= -3 - 2x_3 + 18x_4 + x_5 + x_6 \end{aligned}$$

R_0	$-f_B$	x_1	x_2	x_3	x_4	x_5	x_6	x_7
Before	3	0	0	-2	18	1	1	0

Pivot 3: x_3 enters, x_2 leaves, and $\theta = 0$. Solve the x_2 equation for x_3 , then substitute into every other equation and the objective.

Intuition: After Pivot 3

After substitution: $\mathcal{B} = (x_1, x_3, x_7)$.

$$\begin{aligned} x_1 &= \frac{64}{3}x_2 + 4x_4 + \frac{4}{3}x_5 - \frac{8}{3}x_6 \\ x_3 &= -\frac{8}{3}x_2 + 10x_4 + \frac{4}{3}x_5 - \frac{2}{3}x_6 \\ x_7 &= 1 + \frac{8}{3}x_2 - 10x_4 - \frac{4}{3}x_5 + \frac{2}{3}x_6 \\ f &= -3 + \frac{16}{3}x_2 - 2x_4 - \frac{5}{3}x_5 + \frac{7}{3}x_6 \end{aligned}$$

R_0	$-f_B$	x_1	x_2	x_3	x_4	x_5	x_6	x_7
After	3	0	$\frac{16}{3}$	0	-2	$-\frac{5}{3}$	$\frac{7}{3}$	0

Same point, same $f = -3$. The first change in R_0 is $0 \rightarrow \frac{16}{3}$ in the x_2 column: strict lexicographic progress, even with a zero step.

Intuition: Strict Progress Prevents Repeated Bases

With a lex-positive start, the rule preserves lex-positive constraint rows and makes R_0 strictly increase at every pivot.

A basis determines its objective row

For the same LP and fixed column order, returning to a basis would reproduce exactly the same R_0 .

The row strictly increases at every pivot

A strictly increasing lexicographic sequence cannot return to an earlier row. Therefore no basis can repeat.

There are only finitely many bases

Simplex must stop: either all reduced costs are nonnegative, or an improving column has no eligible leaving row. These certify optimality or unboundedness, respectively.

The complete algebraic proof is available in the appendix, and textbook Theorem 3.4.

Make the Initial Constraint Rows Lex Positive

The usual column order already worked for Example 3.6. For a general feasible basis B_0 , an easy setup is to put its initial basic columns first and then keep that order fixed.

$$\left[\begin{array}{c|c|c} B_0^{-1}b & I & B_0^{-1}A_{\mathcal{N}_0} \end{array} \right].$$

If $\bar{b}_i > 0$, the first entry already makes row i positive.

If $\bar{b}_i = 0$, its first nonzero entry in the identity block is $+1$. Entries in the later nonbasic block cannot change that comparison.

Every row starts lexicographically positive. Theorem 3.4 then preserves this property through all subsequent pivots.

Basic Columns First: A Convenient Setup

Restart Example 3.6 at (x_5, x_6, x_7) . Apply the setup just described and fix the comparison order:

$$\text{RHS} \mid x_5, x_6, x_7 \mid x_1, x_2, x_3, x_4.$$

R_{x_i} is the row for basic x_i ; divide by its x_1 -column entry.

row	RHS	x_5	x_6	x_7	x_1	x_2	x_3	x_4
$R_{x_5}/(\frac{1}{4})$	0	4	0	0	1	−32	−4	36
$R_{x_6}/(\frac{1}{2})$	0	0	2	0	1	−24	−1	6

x_1 still enters. The RHS entries tie; the very next entry gives $0 < 4$. x_6 **leaves at step zero.**

The identity block guarantees a lex-positive start. Here it also resolves the tie one entry earlier than the usual column order.

One Pivot with Basic Columns First

Solve the initial x_6 equation for x_1 , then substitute into the other equations and the objective:

$$x_1 = 24x_2 + x_3 - 6x_4 - 2x_6$$

$$x_5 = 2x_2 + \frac{3}{4}x_3 - \frac{15}{2}x_4 + \frac{1}{2}x_6$$

$$x_7 = 1 - x_3$$

$$f = -3 + 2x_2 - \frac{5}{4}x_3 + \frac{21}{2}x_4 + \frac{3}{2}x_6$$

New basis: (x_1, x_5, x_7) . The point is unchanged and $f = -3$. Keep the comparison order fixed if continuing this run.

Different fixed orders can give different pivot paths. This setup guarantees the starting row condition; it does not guarantee fewer pivots.

From Row Comparisons to Variable Indices

Lexicographic comparison prevents cycling. Can we also prevent cycling using the **variable indices**?

Lex: look farther along the rows

When the ordinary minimum ratios tie, compare subsequent entries of the normalized rows in the fixed column order.

Bland: use the basic-variable indices

Among minimum-ratio ties, let the basic variable with the smallest index leave. If x_8 and x_3 tie, choose x_3 , regardless of row position.

The leaving choice alone is not enough. Bland also prescribes the entering choice. We need both parts of the rule for its termination guarantee.

Bland's Rule: Use the Smallest Variable Indices

1 • Entering variable

Among nonbasic variables with $\bar{c}_j < 0$, choose the smallest variable index j .

2 • Leaving variable

Perform the ordinary minimum-ratio test. Among its ties, choose the smallest index of the basic variable that can leave.

Variable index is not row position. If rows for x_8 and x_3 tie, choose x_3 , even when it appears lower in the tableau.

Why Bland’s Rule Does Not Repeat Our Cycle

The first four entering choices agree. At basis (x_3, x_4, x_7) , the rules separate:

Tableau where Bland’s entering choice breaks the six-pivot cycle

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	3	$-\frac{1}{2}$	16	0	0	−1	1	0
x_3	0	$-\frac{5}{2}$	56	1	0	2	−6	0
x_4	0	$-\frac{1}{4}$	$\frac{16}{3}$	0	1	$\frac{1}{3}$	$-\frac{2}{3}$	0
x_7	1	$\frac{5}{2}$	−56	0	0	−2	6	1

$$\bar{c}_1 = -\frac{1}{2}, \quad \bar{c}_5 = -1.$$

Most-negative pricing chooses x_5 . Bland chooses x_1 , the smaller eligible index.

Only x_7 decreases: $\theta^* = 1/(5/2) = 2/5$. The objective drops to $-3 + (2/5)(-1/2) = -16/5$.

Bland's Rule Guarantees Finite Termination

Termination theorem

Starting from a feasible basis, simplex with Bland's entering and leaving rules never cycles and terminates after finitely many pivots.

Both choices matter. Our cycling example already used the smallest-index leaving variable, but its most-negative entering rule still cycled.

Finite Termination Gives an Optimal Basis

Assumptions

The standard-form LP is feasible and has a finite optimal value. Start from a feasible basis and use an anticycling rule.

Conclusion

Unboundedness is ruled out, so simplex terminates at a basis satisfying

$$\underbrace{B^{-1}b \geq 0}_{\text{feasible}} \quad \text{and} \quad \underbrace{\bar{c}_{\mathcal{N}} \geq 0}_{\text{optimality condition}} .$$

The resulting basis is optimal—even if it is degenerate.

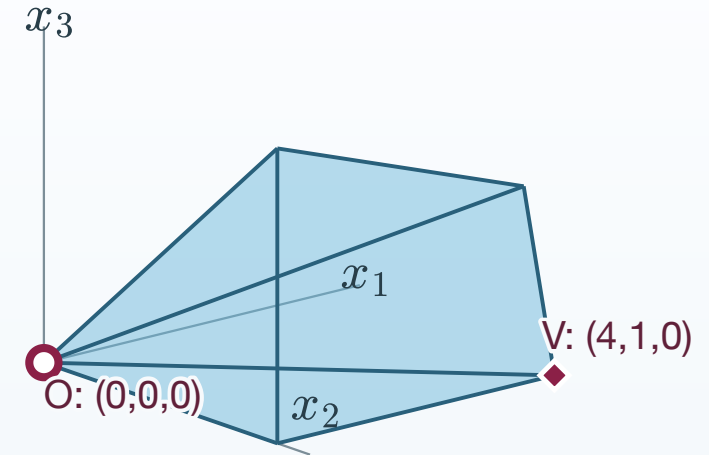
What do these basis changes look like in three dimensions?

Compare the Rules on a Three-Dimensional LP

A zero-length pivot changes the basis. Can several such pivots return to the starting basis?

$$\begin{aligned} \min \quad & f = -2x_1 - 2x_2 + 9x_3 \\ \text{s.t.} \quad & 0.6x_1 - 6.4x_2 + 4.8x_3 \leq 0, \\ & 0.2x_1 - 1.8x_2 + 0.6x_3 \leq 0, \\ & 0.4x_1 - 1.6x_2 + 0.2x_3 \leq 0, \\ & x_2 \leq 1, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Start at the apex $O = (0, 0, 0)$, with $f = 0$. Several constraints are tight there.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

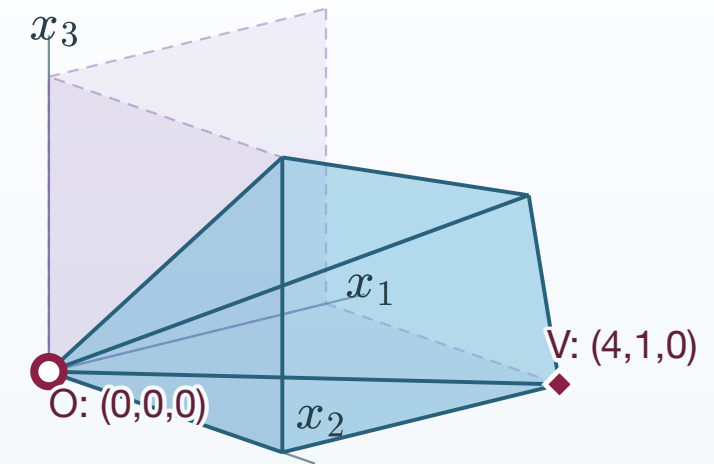
A Cycle of Bases at One Fixed Point

Each row lists basic-variable indices. Every row represents the same point O, with $f = 0$.

After pivot	Basis indices
0	4, 5, 6, 7
1	1, 5, 6, 7
2	1, 2, 6, 7
3	1, 2, 3, 7
4	2, 3, 4, 7
5	3, 4, 5, 7
6	4, 5, 6, 7

After six zero-length pivots, the initial dictionary returns.
The same choices repeat forever.

An anticycling rule must prevent repeated bases.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

The geometry stays fixed while the basis changes.

The Lex Rule Breaks the 3D Cycle at Pivot 4

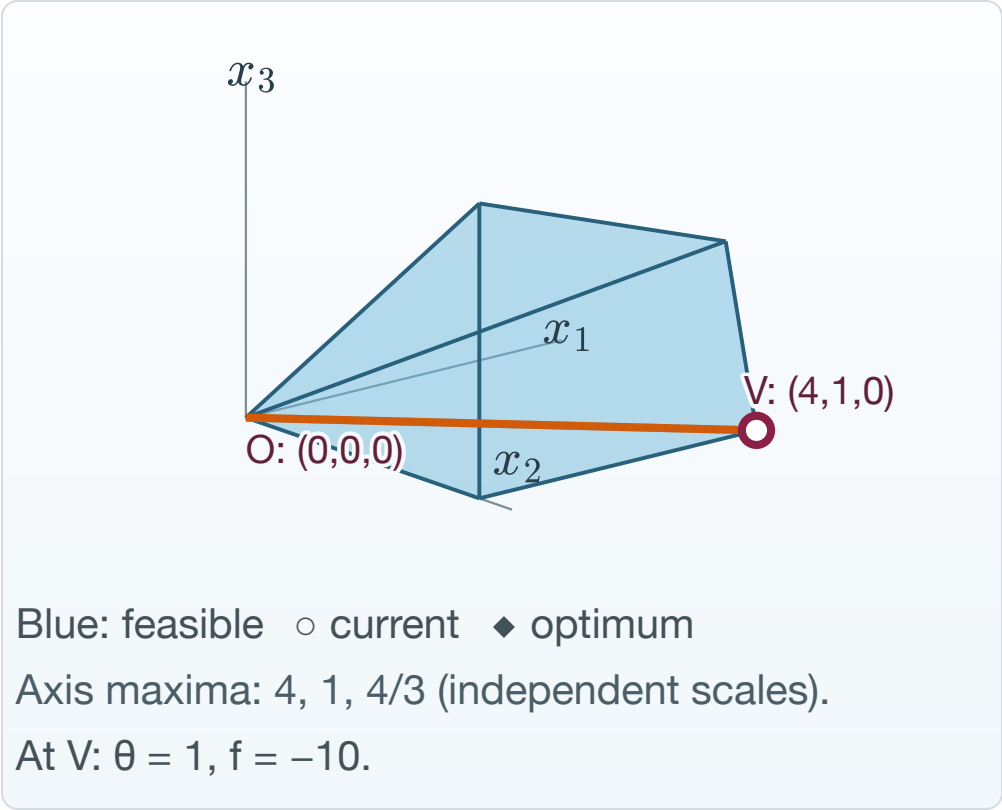
Use RHS, then $x_1, \dots, x_7$. The first three pivots match the cycling run; now the basis is (x_1, x_2, x_3, x_7) .

x_4 enters. These normalized-row prefixes settle the zero-ratio tie:

row	RHS	x_1	x_2
$R_{x_1}/(0.6)$	0	$\frac{5}{3}$	0
$R_{x_2}/(0.2)$	0	0	5
$R_{x_3}/(0.4)$	0	0	0

x_3 **leaves** at $\theta = 0$. Its prefix is smallest. The cycling rule chose x_1 .

Pivot 5: x_5 enters, x_7 leaves at $\theta = 1$. We reach $(4, 1, 0)$, with $f = -10$.



Bland Changes the Sixth Pivot

After five pivots, compare the entering choices:

$$x_3 = -2x_1 + 8x_2 - 5x_6$$

$$x_4 = 9x_1 - 32x_2 + 24x_6$$

$$x_5 = x_1 - 3x_2 + 3x_6$$

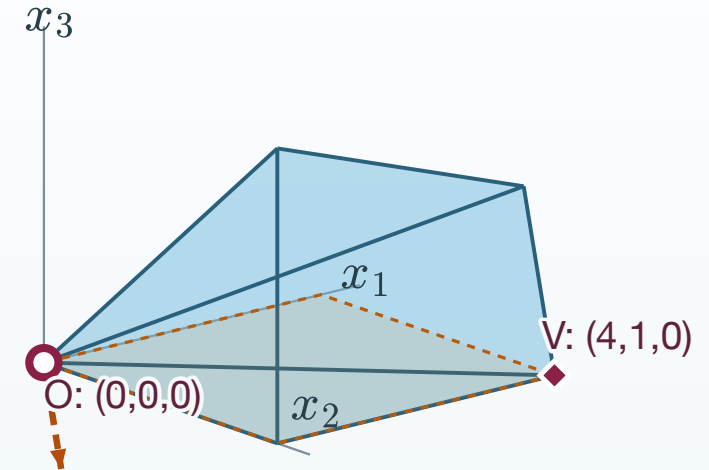
$$x_7 = 1 - x_2$$

$$f = -20x_1 + 70x_2 - 45x_6$$

$$\bar{c}_1 = -20, \quad \bar{c}_6 = -45.$$

Most-negative pricing chooses x_6 . Bland chooses the smaller eligible index, x_1 .

x_1 **enters**; x_3 **leaves at** $\theta = 0$. The new basis is (x_1, x_4, x_5, x_7) . Still O; a different basis.



Dashed: direction (scaled) · orange: blocker
Blue: feasible ○ current ♦ optimum
Axis maxima: 4, 1, 4/3 (independent scales).

The Next Pivot Leaves the Apex

After Bland's sixth pivot:

$$x_1 = 4x_2 - 0.5x_3 - 2.5x_6$$

$$x_4 = 4x_2 - 4.5x_3 + 1.5x_6$$

$$x_5 = x_2 - 0.5x_3 + 0.5x_6$$

$$x_7 = 1 - x_2$$

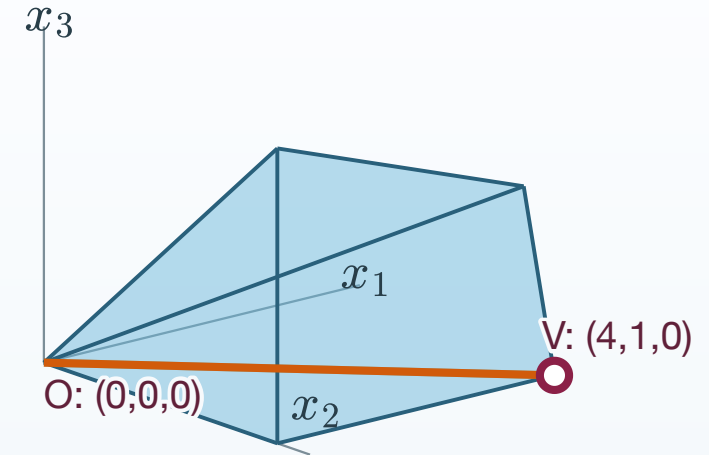
$$f = -10x_2 + 10x_3 + 5x_6$$

Only x_2 has negative reduced cost. Keep $x_3 = x_6 = 0$; set $x_2 = \theta$.

$$x_1 = 4\theta, \quad x_7 = 1 - \theta, \quad 0 \leq \theta \leq 1.$$

At $\theta = 1$, x_7 leaves.

We reach $V = (4, 1, 0)$ with $f = -10$.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

At V: $\theta = 1$, $f = -10$.

Finite Termination Does Not Guarantee Fast Convergence

Lexicographic and Bland rules prevent a basis from repeating. How many *different* bases might we visit?

No repeated bases gives finite termination. It does not promise a small number of pivots.

A long path can consist entirely of positive, improving steps. Slow progress can occur even without degeneracy.

Be precise about the rule: lex chooses the *leaving* row. In the next example, pair it with the most-negative reduced cost entering. Bland chooses both.

Bland has examples with exponentially many pivots as dimension grows. A small 3D example will let us see a detour.

Further reading: Notes on Bland's pivoting rule.

A Short Route Exists, but Pivots Can Take a Detour

Klee–Minty example. Start at $A = (0, 0, 0)$.

min

$f = -4x_1 - 2x_2 - x_3$

s.t.

$x_1 \leq 5,$

$4x_1 + x_2 \leq 25,$

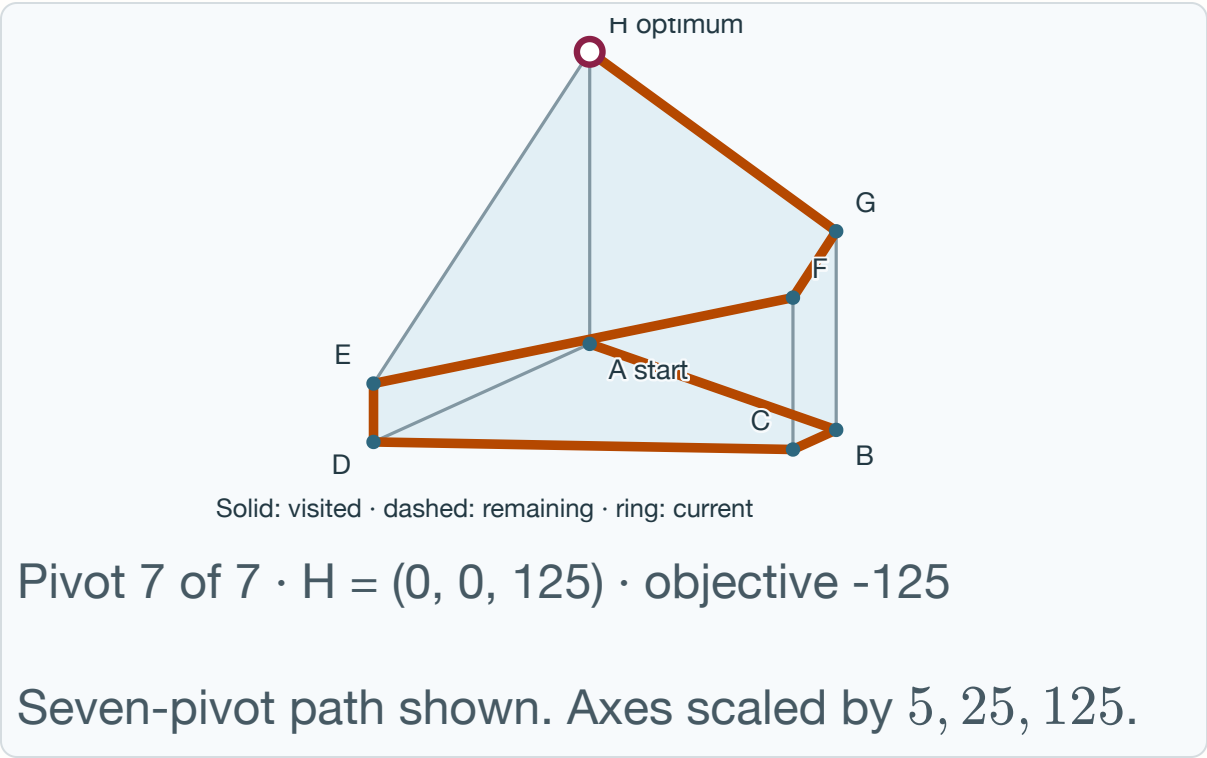
$8x_1 + 4x_2 + x_3 \leq 125,$

$x_1, x_2, x_3 \geq 0.$

Add slacks x_4, x_5, x_6 in row order.

Entering / leaving choice	Pivots
Most-negative / lex	7
Bland / Bland	5
Choose x_3 first	1

Every step is positive: **no degeneracy or cycling.**
 $H = (0, 0, 125), f = -125.$



What Can We Do About Slow Progress?

Choose pivots for practical progress. Simplex solvers use entering-variable strategies such as steepest-edge and Devex: they compare improvement while accounting for direction size. Anticycling safeguards handle degeneracy.

Use a different algorithm family. Interior-point methods are not restricted to edges between vertices. Polynomial-time variants exist; we will study the idea later.

Neither family is always fastest. Modern solvers offer simplex and interior-point methods.

Next: we can prevent cycling and will return to efficiency in §3.7. First, simplex still needs a starting feasible basis. How do we find one?

[Interior-point foundations](#) · [Choosing an LP algorithm](#)

Review Anticycling and Continue to Simplex III

Check your understanding

Explain why a zero-length pivot can change the basis. Apply lexicographic comparison and Bland's two index choices. Distinguish finite termination from a small pivot count.

Next: find a starting basis

Simplex III: Two-Phase Simplex, Geometry, and Efficiency constructs the first feasible basis, then connects pivots to geometry and computational work.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §3.4; Theorem 3.4 for the full lexicographic proof. Section 3.7 develops the efficiency questions.

3D example: [Cycling problems in linear programming, Problem 7](#).

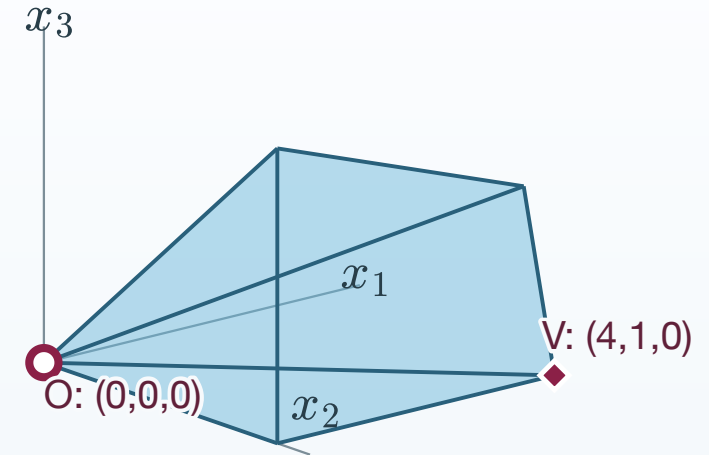
The appendix contains the complete 3D dictionaries, Example 3.7, and the algebraic proof of Theorem 3.4.

Appendix: A Three-Dimensional Cycling Example

A zero-length pivot changes the basis. Can several such pivots return to the starting basis?

$$\begin{aligned} \min \quad & f = -2x_1 - 2x_2 + 9x_3 \\ \text{s.t.} \quad & 0.6x_1 - 6.4x_2 + 4.8x_3 \leq 0, \\ & 0.2x_1 - 1.8x_2 + 0.6x_3 \leq 0, \\ & 0.4x_1 - 1.6x_2 + 0.2x_3 \leq 0, \\ & x_2 \leq 1, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Start at the apex $O = (0, 0, 0)$, with $f = 0$. Several constraints are tight there.



Blue: feasible $\circ$ current $\blacklozenge$ optimum

Axis maxima: 4, 1, $4/3$ (independent scales).

Start with the Slack Dictionary

Add nonnegative slacks x_4, x_5, x_6, x_7 , in constraint order.

$$x_4 = -0.6x_1 + 6.4x_2 - 4.8x_3$$

$$x_5 = -0.2x_1 + 1.8x_2 - 0.6x_3$$

$$x_6 = -0.4x_1 + 1.6x_2 - 0.2x_3$$

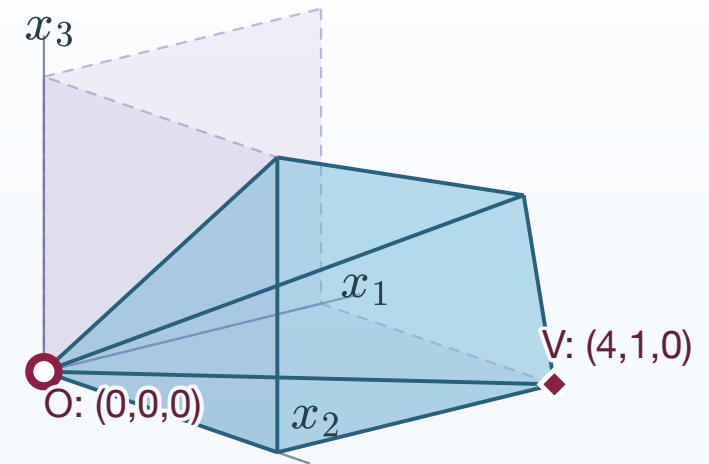
$$x_7 = 1 - x_2$$

$$f = -2x_1 - 2x_2 + 9x_3$$

Slacks have zero cost: f already uses only nonbasic variables.

Set $x_1 = x_2 = x_3 = 0$. Then $(x_4, x_5, x_6, x_7) = (0, 0, 0, 1)$.

Three basic variables are zero: this basis is **degenerate**.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

A tight plane may touch the pyramid only at O.

From Dictionary Equations to a Tableau

A **tableau** records right-hand sides and left-hand coefficients. Columns and basic-variable rows follow increasing variable index.

$$x_4 = -0.6x_1 + 6.4x_2 - 4.8x_3 \iff 0.6x_1 - 6.4x_2 + 4.8x_3 + x_4 = 0.$$

$$f = -2x_1 - 2x_2 + 9x_3 \iff -f - 2x_1 - 2x_2 + 9x_3 = 0.$$

row	column 0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
R_0	0	-2	-2	9	0	0	0	0
x_4	0	0.6	-6.4	4.8	1	0	0	0
x_5	0	0.2	-1.8	0.6	0	1	0	0
x_6	0	0.4	-1.6	0.2	0	0	1	0
x_7	1	0	1	0	0	0	0	1

R_0 omits the $-f$ coefficient. Column 0 stores $-f_B = 0$; the remaining entries are reduced costs.

3D Cycling: Pivot 1

Choose the most negative reduced cost. Break ties by variable index.

$$x_4 = -0.6x_1 + 6.4x_2 - 4.8x_3$$

$$x_5 = -0.2x_1 + 1.8x_2 - 0.6x_3$$

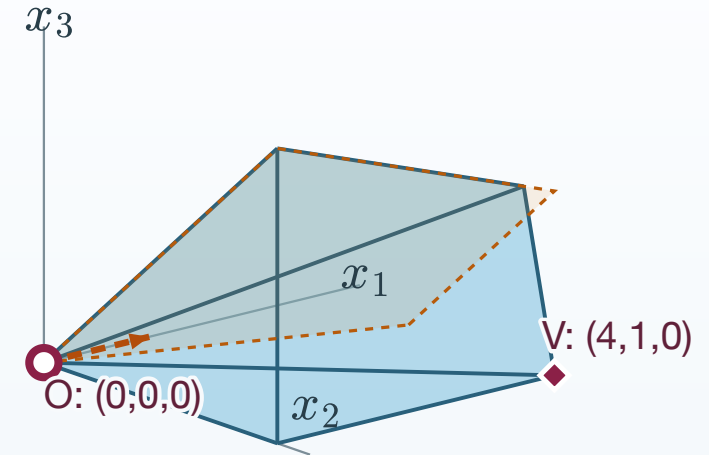
$$x_6 = -0.4x_1 + 1.6x_2 - 0.2x_3$$

$$x_7 = 1 - x_2$$

$$f = -2x_1 - 2x_2 + 9x_3$$

$$\bar{c}_1 = -2, \quad \theta^* = \min \left\{ \frac{0}{0.6}, \frac{0}{0.2}, \frac{0}{0.4} \right\} = 0.$$

$\bar{c}_1 = \bar{c}_2 = -2$: choose x_1 by index. But $x_4 = -0.6\theta \geq 0$ forces $\theta = 0$. Three ratios tie; x_4 leaves by index.



Dashed: direction (scaled) · orange: blocker
Blue: feasible ○ current ♦ optimum
Axis maxima: 4, 1, 4/3 (independent scales).

Nonbasic: x_1, x_2, x_3 .

x_1 **enters**; x_4 **leaves**.

3D Cycling: Pivot 2

After pivot 1: same point $O = (0, 0, 0)$, same $f = 0$.

$$x_1 = \frac{32}{3}x_2 - 8x_3 - \frac{5}{3}x_4$$

$$x_5 = -\frac{1}{3}x_2 + x_3 + \frac{1}{3}x_4$$

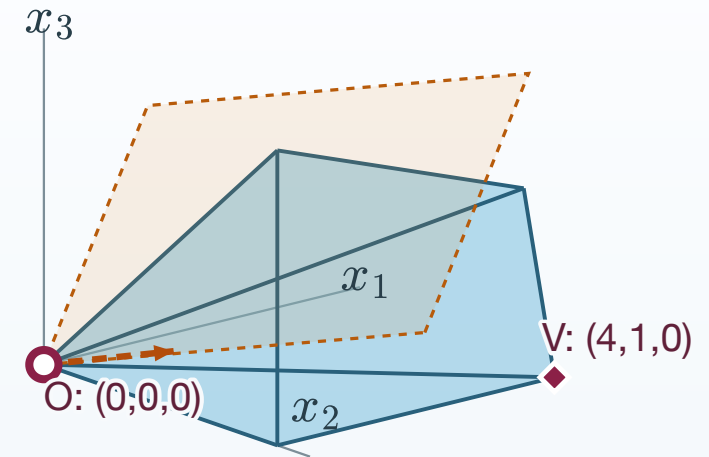
$$x_6 = -\frac{8}{3}x_2 + 3x_3 + \frac{2}{3}x_4$$

$$x_7 = 1 - x_2$$

$$f = -\frac{70}{3}x_2 + 25x_3 + \frac{10}{3}x_4$$

$$\bar{c}_2 = -\frac{70}{3}, \quad \theta^* = \min\{0, 0, 1\} = 0.$$

Zero tie: x_5 leaves before x_6 by index.



Dashed: direction (scaled) · orange: blocker
Blue: feasible ○ current ♦ optimum
Axis maxima: 4, 1, 4/3 (independent scales).

Nonbasic: x_2, x_3, x_4 .

x_2 **enters**; x_5 **leaves**.

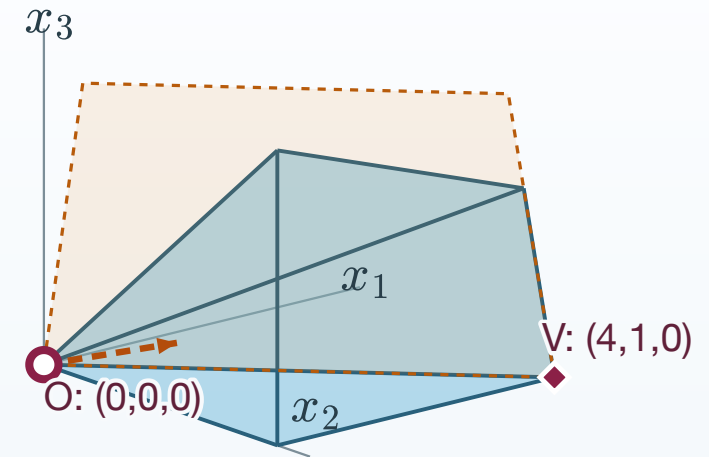
3D Cycling: Pivot 3

After pivot 2: same point $O = (0, 0, 0)$, same $f = 0$.

$$\begin{aligned}x_1 &= 24x_3 + 9x_4 - 32x_5 \\x_2 &= 3x_3 + x_4 - 3x_5 \\x_6 &= -5x_3 - 2x_4 + 8x_5 \\x_7 &= 1 - 3x_3 - x_4 + 3x_5 \\f &= -45x_3 - 20x_4 + 70x_5\end{aligned}$$

$$\bar{c}_3 = -45, \quad \theta^* = \min\{0, \frac{1}{3}\} = 0.$$

x_6 is zero and would decrease. It blocks the move before the x_7 bound of $1/3$.



Dashed: direction (scaled) · orange: blocker
Blue: feasible ○ current ♦ optimum
Axis maxima: 4, 1, $4/3$ (independent scales).

Nonbasic: x_3, x_4, x_5 .

x_3 enters; x_6 leaves.

3D Cycling: Pivot 4

After pivot 3: same point $O = (0, 0, 0)$, same $f = 0$.

$$x_1 = -0.6x_4 + 6.4x_5 - 4.8x_6$$

$$x_2 = -0.2x_4 + 1.8x_5 - 0.6x_6$$

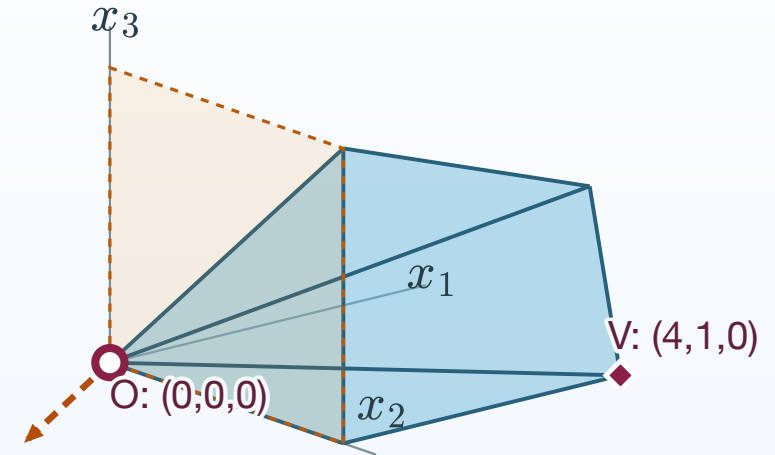
$$x_3 = -0.4x_4 + 1.6x_5 - 0.2x_6$$

$$x_7 = 1 + 0.2x_4 - 1.8x_5 + 0.6x_6$$

$$f = -2x_4 - 2x_5 + 9x_6$$

$$\bar{c}_4 = -2, \quad \theta^* = \min\{0, 0, 0\} = 0.$$

$\bar{c}_4 = \bar{c}_5 = -2$: choose x_4 . All three zero ratios tie; x_1 leaves.



Dashed: direction (scaled) · orange: blocker

Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

Nonbasic: x_4, x_5, x_6 .

x_4 enters; x_1 leaves.

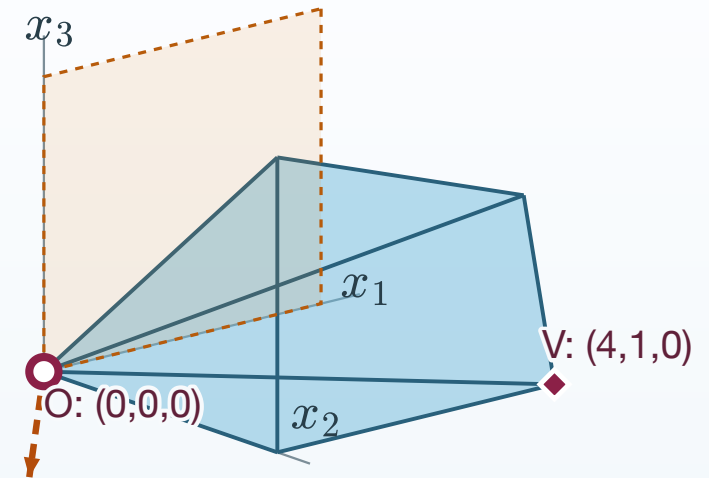
3D Cycling: Pivot 5

After pivot 4: same point $O = (0, 0, 0)$, same $f = 0$.

$$\begin{aligned}x_2 &= \frac{1}{3}x_1 - \frac{1}{3}x_5 + x_6 \\x_3 &= \frac{2}{3}x_1 - \frac{8}{3}x_5 + 3x_6 \\x_4 &= -\frac{5}{3}x_1 + \frac{32}{3}x_5 - 8x_6 \\x_7 &= 1 - \frac{1}{3}x_1 + \frac{1}{3}x_5 - x_6 \\f &= \frac{10}{3}x_1 - \frac{70}{3}x_5 + 25x_6\end{aligned}$$

$$\bar{c}_5 = -\frac{70}{3}, \quad \theta^* = \min\{0, 0\} = 0.$$

Zero tie: x_2 leaves before x_3 by index.



Dashed: direction (scaled) · orange: blocker
Blue: feasible ○ current ♦ optimum
Axis maxima: 4, 1, 4/3 (independent scales).

Nonbasic: x_1, x_5, x_6 .

x_5 **enters**; x_2 **leaves**.

3D Cycling: Pivot 6

After pivot 5: same point $O = (0, 0, 0)$, same $f = 0$.

$$x_3 = -2x_1 + 8x_2 - 5x_6$$

$$x_4 = 9x_1 - 32x_2 + 24x_6$$

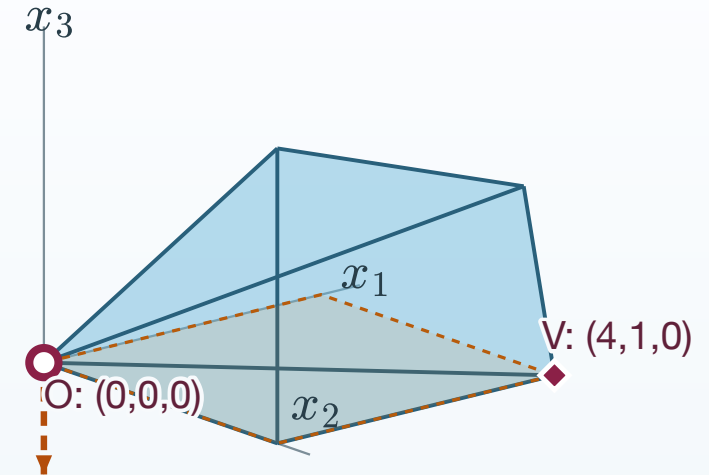
$$x_5 = x_1 - 3x_2 + 3x_6$$

$$x_7 = 1 - x_2$$

$$f = -20x_1 + 70x_2 - 45x_6$$

$$\bar{c}_6 = -45, \quad \theta^* = \min\{0\} = 0.$$

$\bar{c}_6 = -45 < -20 = \bar{c}_1$: choose x_6 . Only x_3 limits its increase.



Dashed: direction (scaled) · orange: blocker

Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

Nonbasic: x_1, x_2, x_6 .

x_6 **enters**; x_3 **leaves**.

After Pivot 6: The Initial Dictionary Returns

x_6 entered and x_3 left. Compare this with our first dictionary:

$$x_4 = -0.6x_1 + 6.4x_2 - 4.8x_3$$

$$x_5 = -0.2x_1 + 1.8x_2 - 0.6x_3$$

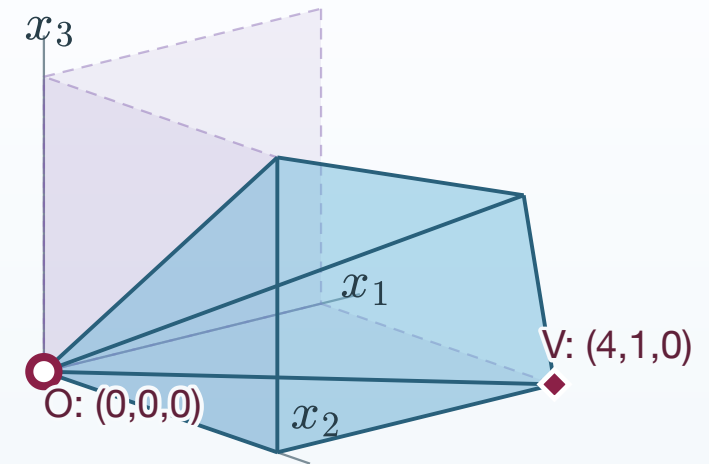
$$x_6 = -0.4x_1 + 1.6x_2 - 0.2x_3$$

$$x_7 = 1 - x_2$$

$$f = -2x_1 - 2x_2 + 9x_3$$

Every coefficient is restored. The basis is (x_4, x_5, x_6, x_7) , and the point is still O.

The same rules choose x_1 again. The entire six-pivot sequence repeats.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

Six basis changes. No displacement. No change in $f = 0$.

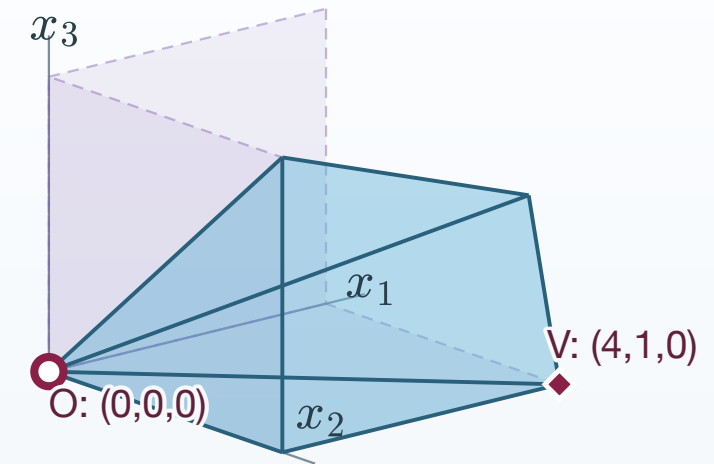
A Cycle of Bases at One Fixed Point

Each row lists basic-variable indices. Every row represents the same point O, with $f = 0$.

After pivot	Basis indices
0	4, 5, 6, 7
1	1, 5, 6, 7
2	1, 2, 6, 7
3	1, 2, 3, 7
4	2, 3, 4, 7
5	3, 4, 5, 7
6	4, 5, 6, 7

After six zero-length pivots, the initial dictionary returns.
The same choices repeat forever.

An anticycling rule must prevent repeated bases.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

The geometry stays fixed while the basis changes.

The Lex Rule Breaks the 3D Cycle at Pivot 4

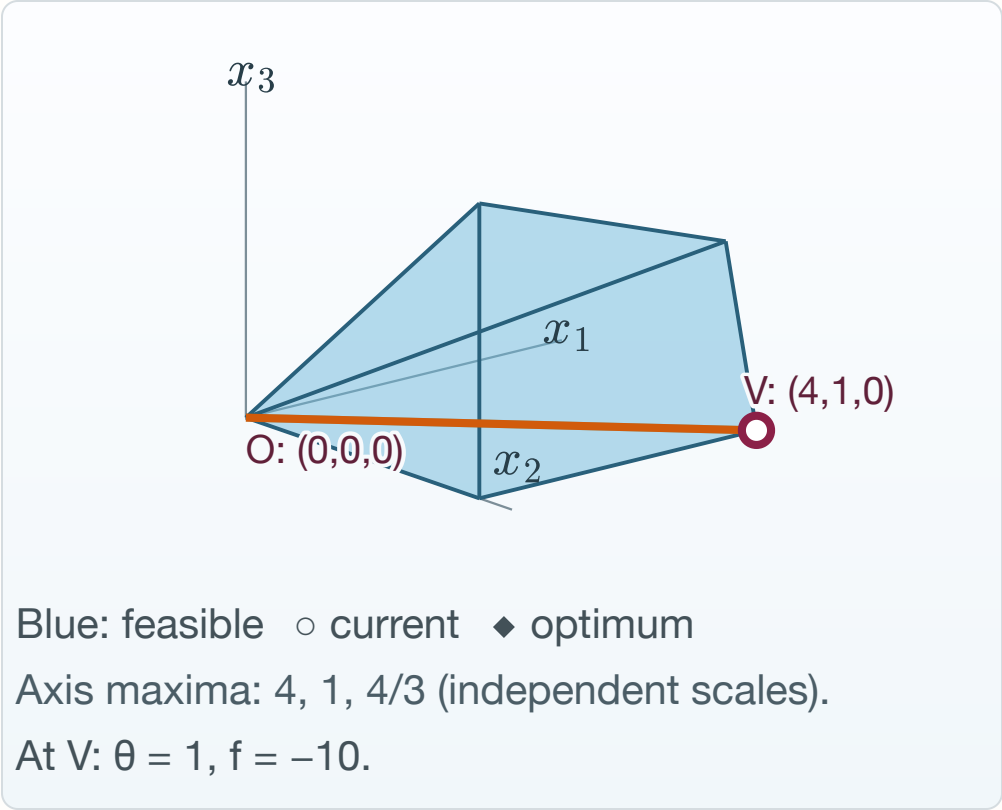
Use RHS, then $x_1, \dots, x_7$. The first three pivots match the cycling run; now the basis is (x_1, x_2, x_3, x_7) .

x_4 enters. These normalized-row prefixes settle the zero-ratio tie:

row	RHS	x_1	x_2
$R_{x_1}/(0.6)$	0	$\frac{5}{3}$	0
$R_{x_2}/(0.2)$	0	0	5
$R_{x_3}/(0.4)$	0	0	0

x_3 **leaves** at $\theta = 0$. Its prefix is smallest. The cycling rule chose x_1 .

Pivot 5: x_5 enters, x_7 leaves at $\theta = 1$. We reach $(4, 1, 0)$, with $f = -10$.



Bland Changes the Sixth Pivot

After five pivots, compare the entering choices:

$$x_3 = -2x_1 + 8x_2 - 5x_6$$

$$x_4 = 9x_1 - 32x_2 + 24x_6$$

$$x_5 = x_1 - 3x_2 + 3x_6$$

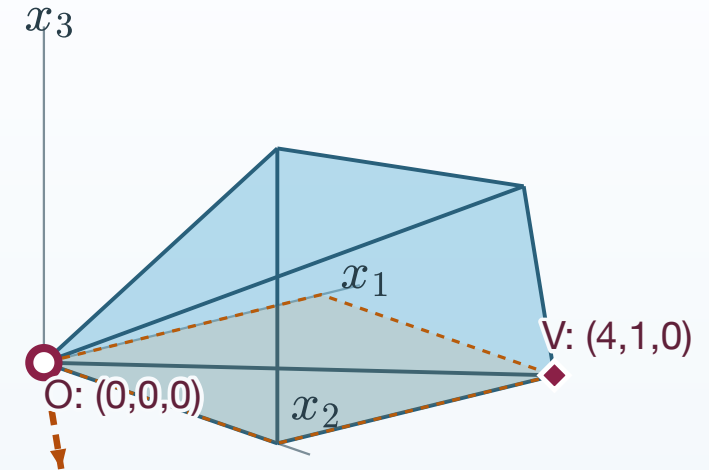
$$x_7 = 1 - x_2$$

$$f = -20x_1 + 70x_2 - 45x_6$$

$$\bar{c}_1 = -20, \quad \bar{c}_6 = -45.$$

Most-negative pricing chooses x_6 . Bland chooses the smaller eligible index, x_1 .

x_1 **enters**; x_3 **leaves at** $\theta = 0$. The new basis is (x_1, x_4, x_5, x_7) . Still O; a different basis.



Dashed: direction (scaled) · orange: blocker
Blue: feasible ○ current ♦ optimum
Axis maxima: 4, 1, 4/3 (independent scales).

The Next Pivot Leaves the Apex

After Bland's sixth pivot:

$$x_1 = 4x_2 - 0.5x_3 - 2.5x_6$$

$$x_4 = 4x_2 - 4.5x_3 + 1.5x_6$$

$$x_5 = x_2 - 0.5x_3 + 0.5x_6$$

$$x_7 = 1 - x_2$$

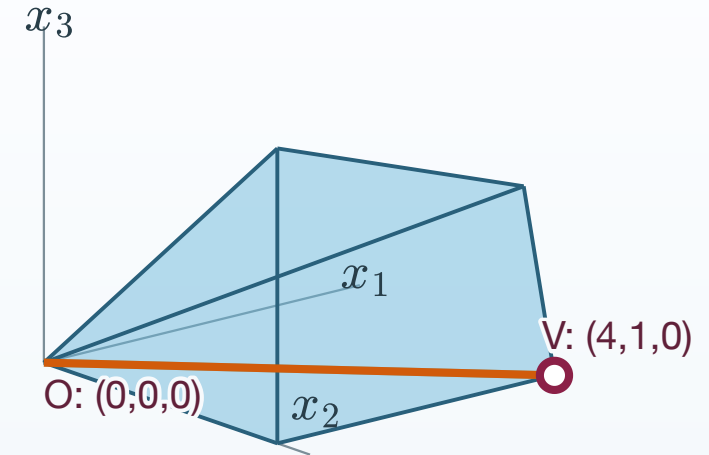
$$f = -10x_2 + 10x_3 + 5x_6$$

Only x_2 has negative reduced cost. Keep $x_3 = x_6 = 0$; set $x_2 = \theta$.

$$x_1 = 4\theta, \quad x_7 = 1 - \theta, \quad 0 \leq \theta \leq 1.$$

At $\theta = 1$, x_7 leaves.

We reach $V = (4, 1, 0)$ with $f = -10$.



Blue: feasible ○ current ♦ optimum

Axis maxima: 4, 1, 4/3 (independent scales).

At V: $\theta = 1$, $f = -10$.

Example 3.7: The Ordinary Ratios Tie

The objective row is omitted. Column 0 holds the basic values $\bar{b}_i$. Let column $j = 3$ enter.

Original tableau rows

basic	column 0	column 1	column 2	column 3	remaining
$x_{B(1)}$	1	0	5	3	...
$x_{B(2)}$	2	4	6	−1	...
$x_{B(3)}$	3	0	7	9	...

$$\frac{\bar{b}_1}{t_{13}} = \frac{1}{3}, \quad \frac{\bar{b}_3}{t_{33}} = \frac{3}{9} = \frac{1}{3}.$$

Row 2 is ineligible because $t_{23} = -1$: that basic variable increases. Rows 1 and 3 tie for the maximum feasible step.

Example 3.7: Compare the Complete Normalized Rows

Divide row 1 by 3 and row 3 by 9. Keep column 0 at the left.

Normalized eligible rows					
row	column 0	column 1	column 2	column 3	remaining
$R_1/3$	$1/3$	0	$5/3$	1	...
$R_3/9$	$1/3$	0	$7/9$	1	...

The ratios tie at $1/3$, and column 1 also ties at zero. Column 2 is the first difference:

$$\frac{7}{9} < \frac{5}{3} \implies \frac{R_3}{9} <_L \frac{R_1}{3}.$$

Row 3 wins: $x_{B(3)}$ leaves and $\theta^* = 1/3$.

Four Lexicographic Facts Used in the Proof

$$r >_L s \iff r - s >_L 0.$$

Subtracting common entries leaves the same first difference.

$$r >_L 0, \alpha > 0 \implies \alpha r >_L 0.$$

Positive scaling preserves the first nonzero sign.

$$r >_L 0, s >_L 0 \implies r + s >_L 0.$$

At the first possible nonzero entry, positive entries cannot cancel.

$$r >_L 0 \implies r + s >_L s.$$

Apply the first fact to $(r + s) - s = r$.

Proof (a): The Pivot Row Stays Positive

Assume all old constraint rows satisfy $R_i >_L 0$. Let x_j enter and let row ℓ leave.

Eligibility gives $t_{\ell j} > 0$. Uniqueness and the lexicographic rule give

$$\frac{R_\ell}{t_{\ell j}} <_L \frac{R_i}{t_{ij}} \quad (i \neq \ell, t_{ij} > 0).$$

$$R'_\ell = \frac{R_\ell}{t_{\ell j}} >_L 0.$$

The new pivot row is a positive multiple of a lexicographically positive row.

The remaining rows use $R'_i = R_i - t_{ij}R'_\ell$. We must check both possible signs of t_{ij} .

Proof (a): Every Other Row Stays Positive

We have $R_i >_L 0$, $R'_\ell = R_\ell/t_{\ell j} >_L 0$, and $R'_i = R_i - t_{ij}R'_\ell$.

Case 1: $t_{ij} \leq 0$

$$R'_i = R_i + (-t_{ij})R'_\ell >_L 0.$$

If $t_{ij} = 0$, the row is unchanged. Otherwise it is the sum of two lexicographically positive vectors.

Case 2: $t_{ij} > 0$

$$R'_i = t_{ij} \left(\frac{R_i}{t_{ij}} - \frac{R_\ell}{t_{\ell j}} \right) >_L 0.$$

The bracket is positive by the strict lexicographic choice; the multiplier is positive.

Induction proves (a): every constraint row is positive initially, and every pivot preserves that condition.

Proof (b): The Zeroth Row Strictly Increases

The entering reduced cost is negative, $\bar{c}_j < 0$, and part (a) gives $R'_\ell >_L 0$.

$$R'_0 = R_0 - \bar{c}_j R'_\ell.$$

This row operation makes the entering reduced cost zero.

$$R'_0 - R_0 = (-\bar{c}_j) R'_\ell >_L 0 \quad \implies \quad R'_0 >_L R_0.$$

Even when $\theta^* = 0$, the whole zeroth row strictly increases. Its first entry $-f_B$ may stay fixed; a later entry then supplies the first strict difference.

Proof (c): A Basis Cannot Recur

For fixed data and column order, the basis determines the complete zeroth row:

$$R_0 = \left[-c_{\mathcal{B}}^{\top} B^{-1} b \mid c^{\top} - c_{\mathcal{B}}^{\top} B^{-1} A \right].$$

Part (b) gives strict lexicographic increase at each pivot. Returning to an old basis would return to its old zeroth row, which is impossible.

There are finitely many choices of basic columns. Therefore the algorithm terminates after finitely many pivots, with an optimal basis or an improving unbounded ray.